

// Exercise 2: E-commerce Platform Search Function

PROGRAM:

```
class Product {
    int productId;
    String productName;
    String category;
    Product(int productId, String productName, String category) {
        this.productId = productId;
        this.productName = productName;
        this.category = category;
    }
}

public class EcommerceSearchFunction {
    static Product linearSearch(Product[] products, int id) {
        for (Product p : products) {
            if (p.productId == id) {
                return p;
            }
        }
        return null;
    }

    static Product binarySearch(Product[] products, int id) {
        int low= 0;
        int high= products.length - 1;
        while (low<= high) {
            int mid = (low + high) / 2;
            if (products[mid].productId == id) {
                return products[mid];
            } else if (products[mid].productId < id) {
                low=mid + 1;
            } else {
                high=mid - 1;
            }
        }
        return null;
    }

    public static void main(String[] args) {
        Product[] products={
```

```

        new Product(1, "Radio", "Electronics"),
        new Product(2, "Television", "Electronics"),
        new Product(3, "Hat", "Fashion"),
        new Product(4, "Watch", "Accessories"),
        new Product(5, "Bag", "Fashion")
    };
    int searchId1=2;
    Product result1 = linearSearch(products, searchId1);
    if (result1 != null) {
        System.out.println("Linear Search Found: " + result1.productName);
    }
    Product result2 = binarySearch(products, searchId1);
    if (result2 != null) {
        System.out.println("Binary Search Found: " + result2.productName);
    }
}
}

```

OUTPUT:

The screenshot shows an IDE with the following components:

- Project View:** Shows a project named 'DSA' with a source folder 'src' containing 'EcommerceSearchFunction.java'.
- Main.java:** Contains the following code:


```

11 public class EcommerceSearchFunction {
12     public static void main(String[] args) {
13         new Product(1, "Radio", "Electronics"),
14         new Product(2, "Television", "Electronics"),
15         new Product(3, "Hat", "Fashion"),
16         new Product(4, "Watch", "Accessories"),
17         new Product(5, "Bag", "Fashion")
18     };
19     int searchId1=2;
20     Product result1 = linearSearch(products, searchId1);
21     if (result1 != null) {
22         System.out.println("Linear Search Found: " + result1.productName);
23     }
24     Product result2 = binarySearch(products, searchId1);
25     if (result2 != null) {
26         System.out.println("Binary Search Found: " + result2.productName);
27     }
28 }
29 }

```
- Run Console:** Shows the output of the program:


```

C:\Users\Naveen\.jdk\openjdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=53671" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Linear Search Found: Television
Binary Search Found: Television
Process finished with exit code 0

```